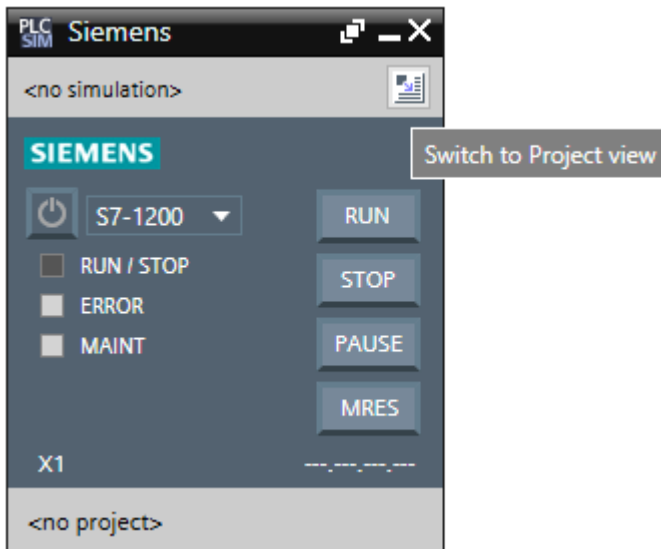


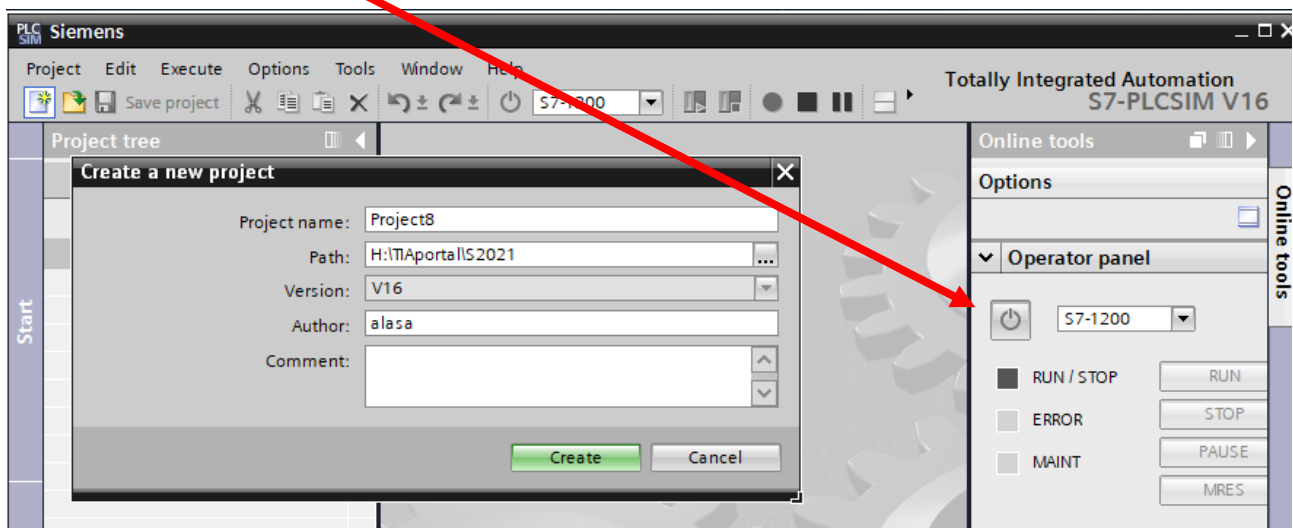
TIA Portal + PLCSim user manual:

1. Open PLCSim and select the project view at the top right

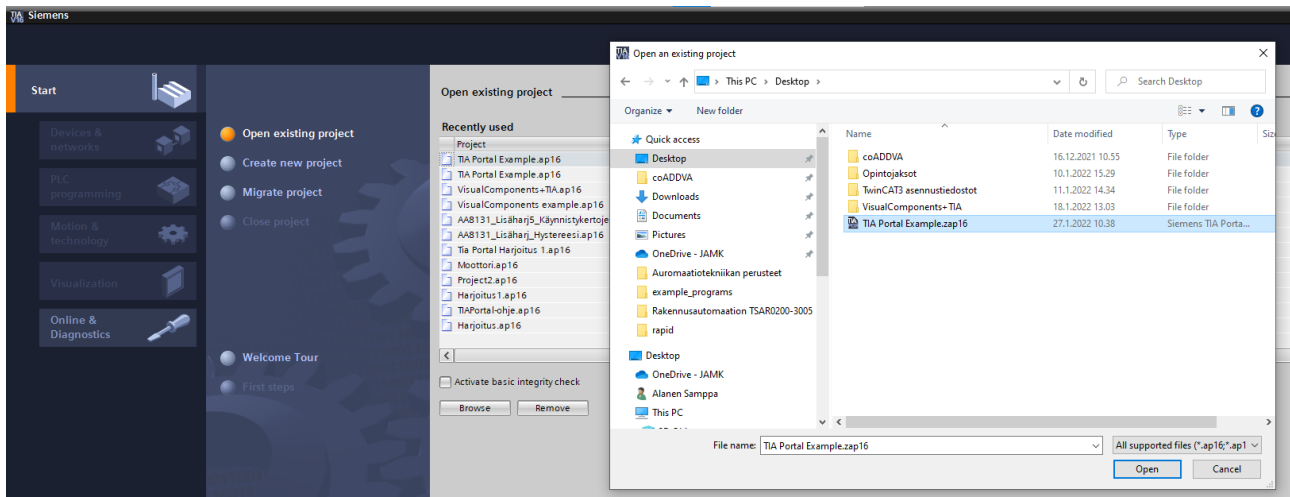


2. Create a new project from the New project button. Give the project a name (no meaning) and select a folder where to save the project (instructions for the H drive, not always allowed to save to drive C). Press **Create**. It will take a while to start. (If you are using Remote connection, save file to the C drive and save the file on the H drive only after you finish)

3. Press the simulator to start. The icon will turn green.

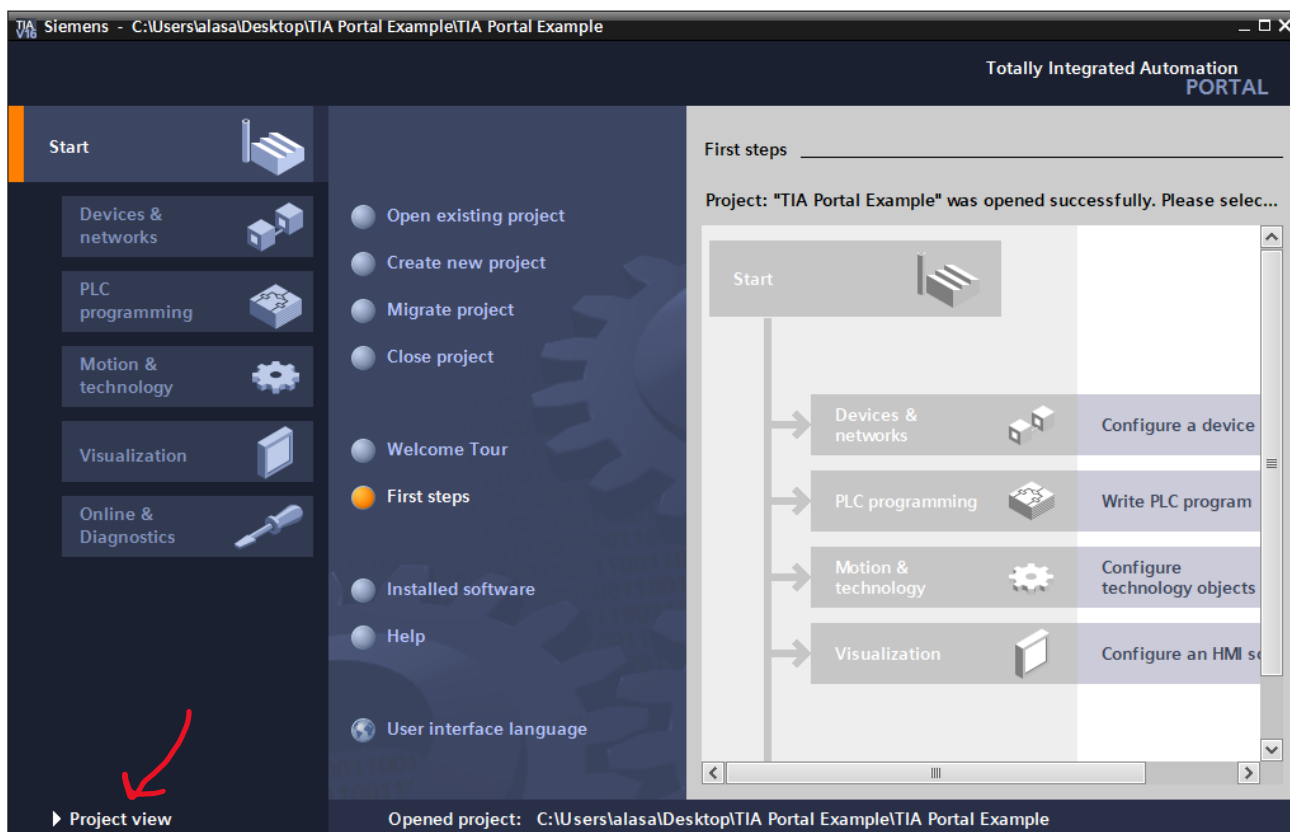


3. Open the TIA Portal.
4. If you have old project; Select the Open existing project > Browse > select the compressed project file xxxx.zap16

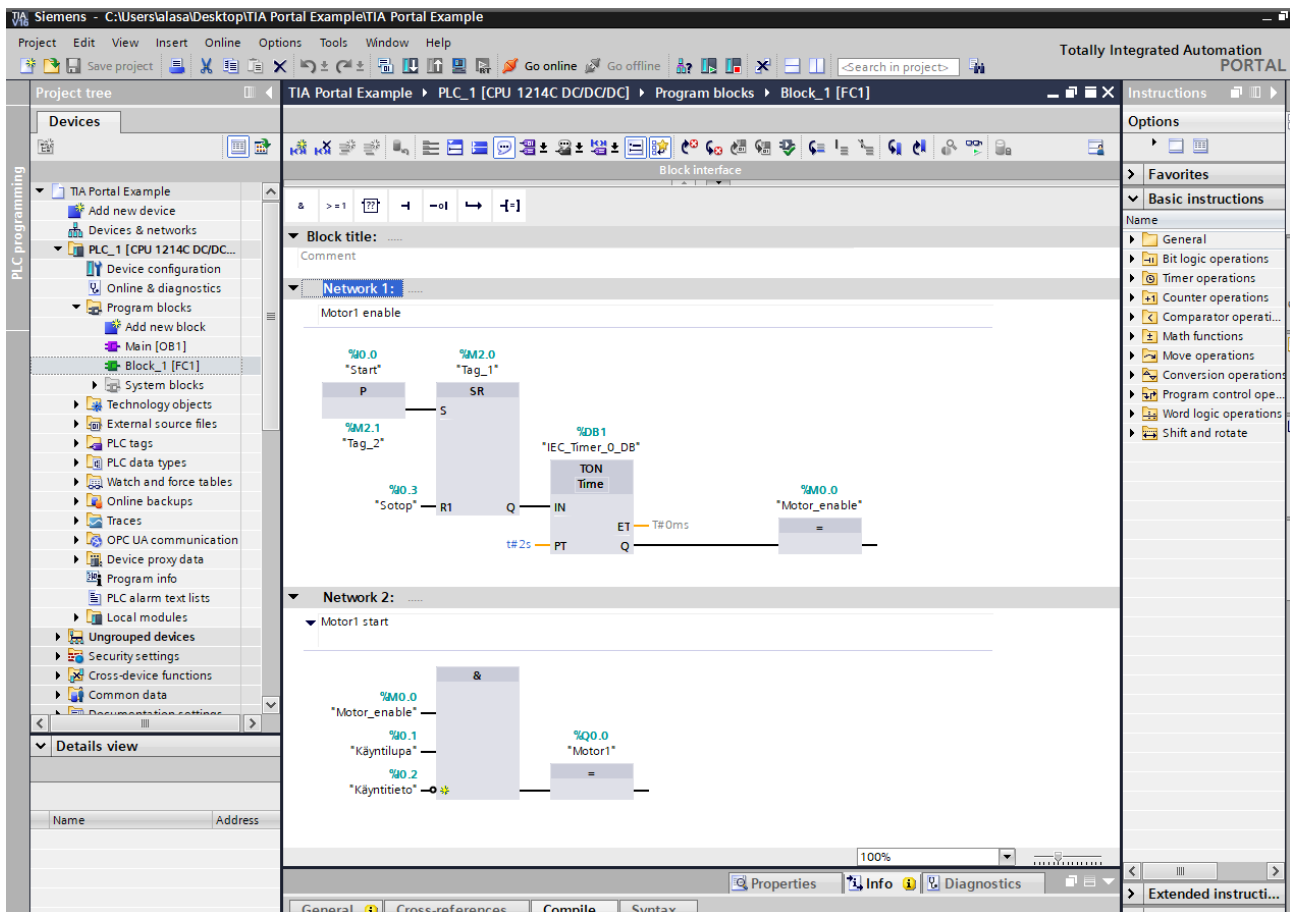


5. Select a save folder for the project (opens the compressed project file).

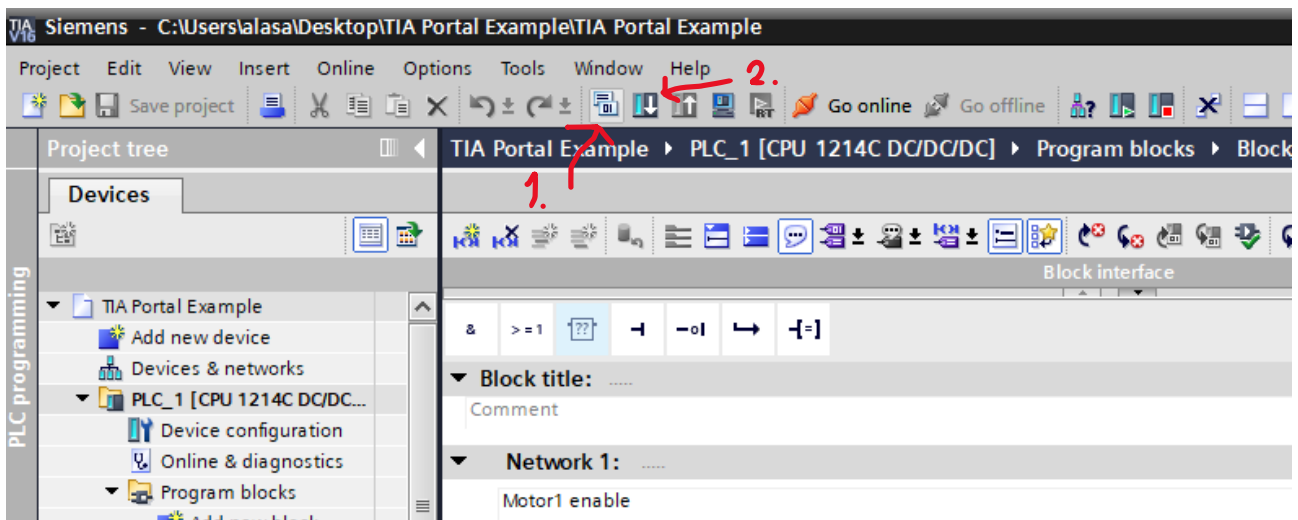
6. At the bottom left of the view, press **Project view**



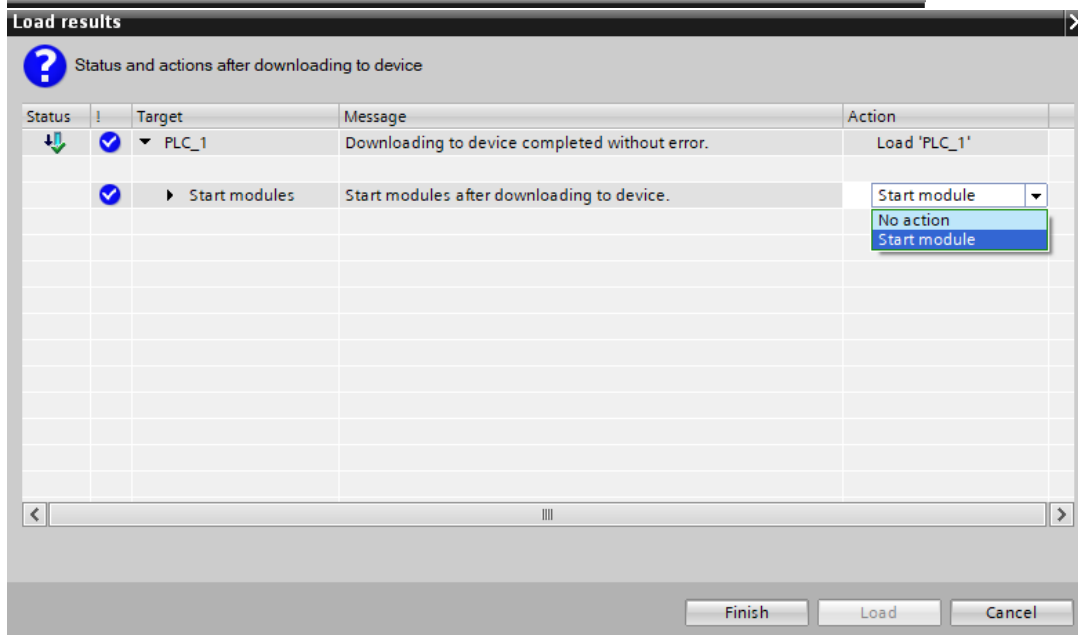
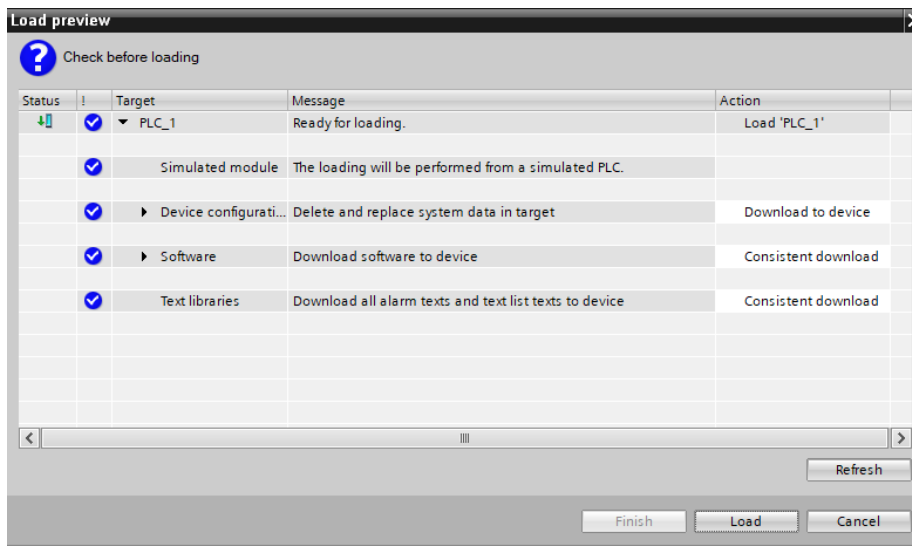
7. In the Project tree view, select the PLC_1 > Program blocks > Block_1 (double-tap)



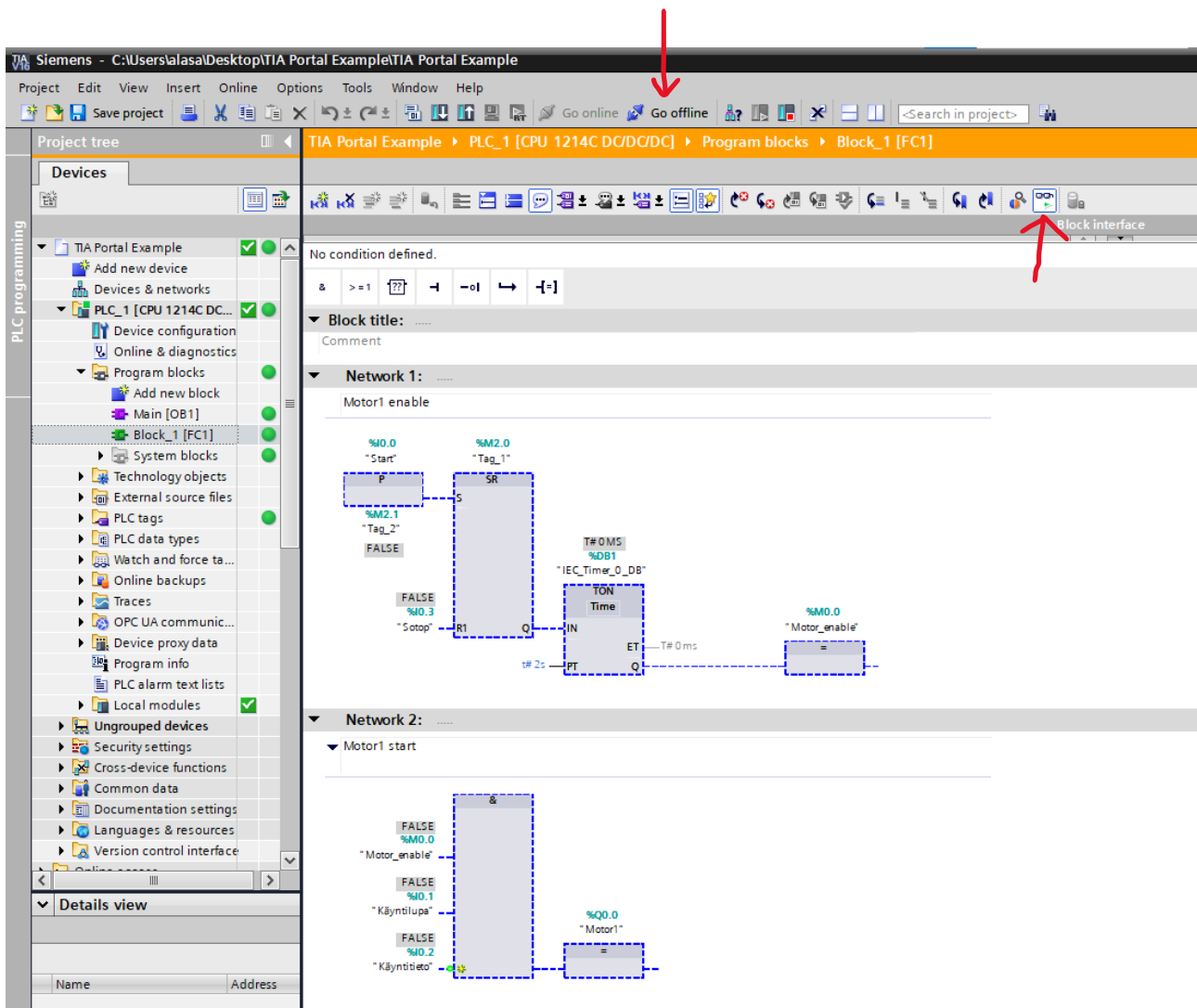
8. Press **Compile** from the top menu of the view. After that, the project can be downloaded to the simulator. Press the button next to **Download to device**.



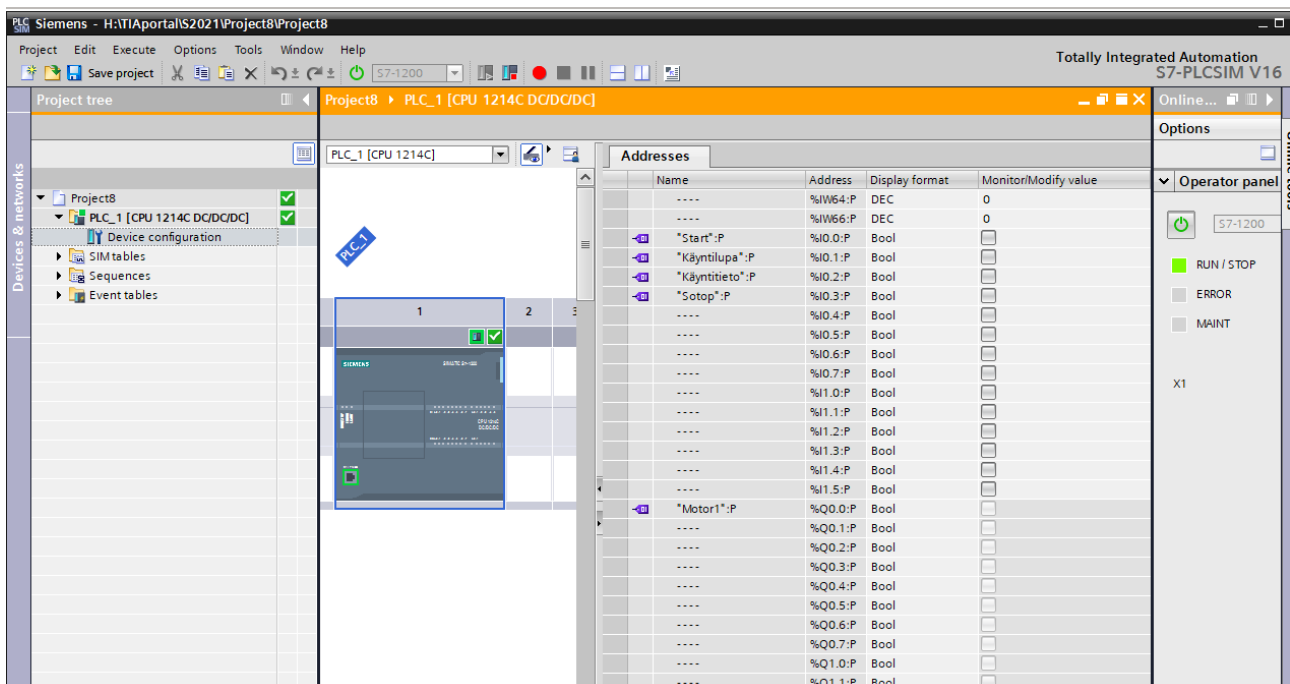
9. Press **Load** and select **Start module**. After that, press **Finish**.



10. At the top, select **Go online** and press **Monitoring on/off** to start



11. Open PLCsim and select the "Project Name" > PLC_1 > Device configuration. Drag the arrows in the center window to open the Addresses view.



12. Arrange the views side by side and test the operation of the program by selecting variables on/off in the Monitor/Modify value table.

- Start and running permission must be on, so that after two seconds the engine1 turns on

The image displays two side-by-side screenshots of the Siemens TIA Portal software interface, showing the Ladder Logic editor and the Addresses table.

Left Window (Ladder Logic Editor):

- Project Tree:** Shows the project structure, including 'TIA Portal Example', 'PLC_1 [CPU 1214C D0D0D0C]', and 'Program blocks'.
- Block Interface:** Displays the Ladder Logic for 'Block 1 [FC1]'.
 - Network 1:** Titled 'Motor1 enable'. It contains a timer TON (S_ODT) set to 2s, triggered by a normally open contact labeled 'Motor1_start'.
 - Network 2:** Titled 'Motor1 start'. It contains a set coil (S) for 'Motor1_start' triggered by a normally open contact labeled 'Motor1_start'.

Right Window (Addresses Table):

- Project Tree:** Shows the project structure, including 'Project8', 'PLC_1 [CPU 1214C D0D0D0C]', and 'Addresses'.
- Addresses Table:** Lists various addresses and their display formats.

Name	Address	Display format	Monitor/Modify
start	%I0.0	Bool	☒
Käynnistys	%I0.1	Bool	☒
Käytöstä poist	%I0.2	Bool	☒
stop	%I0.3	Bool	☒
Motor1	%Q0.0	Bool	☒
Motor2	%Q0.1	Bool	☒
Motor3	%Q0.2	Bool	☒
Motor4	%Q0.3	Bool	☒
Motor5	%Q0.4	Bool	☒
Motor6	%Q0.5	Bool	☒
Motor7	%Q0.6	Bool	☒
Motor8	%Q0.7	Bool	☒
Motor9	%Q0.8	Bool	☒
Motor10	%Q0.9	Bool	☒
Motor11	%Q1.0	Bool	☒
Motor12	%Q1.1	Bool	☒
- Operator panel:** Shows a 'RUN / STOP' button and a 'Motor1' indicator.